

Project Report

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1.INTRODUCTION

1.1 Project Overview:

DocSpot – Seamless Appointment Booking for Health is a full-stack web application designed to simplify and digitize the process of booking medical appointments between patients and healthcare providers. Traditional appointment booking methods often involve long waiting times, manual scheduling, lack of transparency, and inefficient communication between patients and doctors. This project aims to address these challenges by providing a centralized online platform where users can easily search for doctors, check availability, and schedule appointments in real time.

Doctors can manage their schedules, confirm or reschedule appointments, and update consultation details. Admins oversee the overall platform operations, approve doctor registrations, and ensure system compliance. The application follows a modern client-server architecture using React.js for the frontend, Express.js and Node.js for backend services, and MongoDB for data management.

1.2 Purpose:

The primary purpose of the DocSpot application is to enhance accessibility, efficiency, and convenience in healthcare appointment scheduling through a digital solution. The project focuses on reducing manual effort, minimizing waiting times, and improving interaction between patients and doctors.

Key objectives of the system include:

- Providing an intuitive platform for users to discover healthcare professionals and book appointments online.
- Allowing doctors to efficiently manage their schedules and patient requests.
- Enabling administrators to maintain system governance and ensure secure platform operations.
- Delivering real-time appointment updates, notifications, and status tracking.
- Creating a scalable and user-friendly healthcare booking system using modern full-stack technologies.

Overall, the project aims to bridge the gap between patients and healthcare services by offering a reliable and structured appointment management solution that improves user experience and operational efficiency.

2.IDEATION PHASE

2.1 Problem Statement:

In today's healthcare systems, booking and managing doctor appointments can be time-consuming and confusing for both patients and doctors. Many traditional methods do not provide real-time availability or a centralized scheduling system, which leads to missed updates, overlapping requests, and user frustration. The DocSpot platform aims to solve these challenges by providing a streamlined digital solution for efficient appointment booking and management.

Problem Statement -1:

I am	a patient
I'm trying to	book a doctor appointment online quickly
But	I cannot see real-time availability clearly
Because	traditional booking systems don't show updated slots
Which makes me feel	Anxious and frustrated

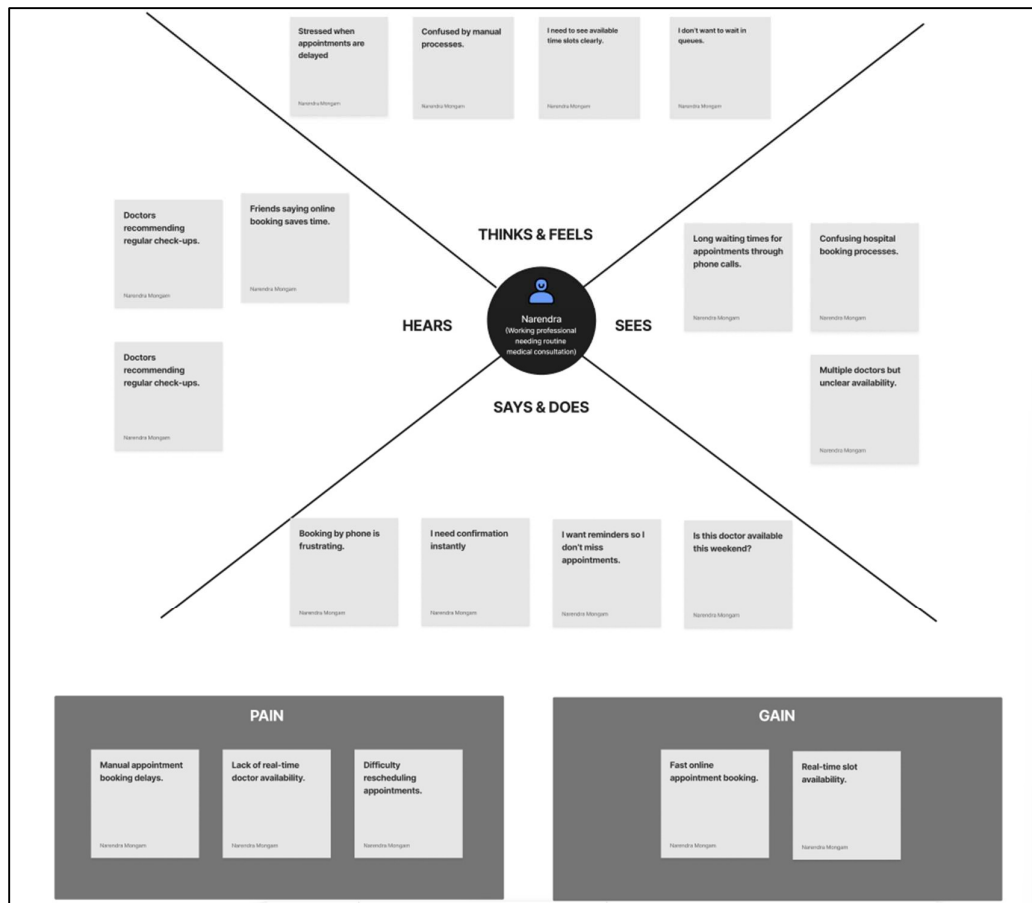
Problem Statement -2:

I am	A doctor
I'm trying to	Manage my daily appointments efficiently
But	Requests come from multiple sources and get confusing
Because	There is no centralized scheduling platform
Which makes me feel	Overwhelmed

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A patient	Book a doctor appointment online quickly	I cannot see real time availability clearly	Traditional booking systems don't show updated slots	Anxious and frustrated
PS-2	A doctor	Manage my daily appointments efficiently	Requests come from multiple sources and get confusing	There is no centralized scheduling platform	Overwhelmed

2.2 Empathy Map:

User: - Narendra Mongam (Working Professional Needs Routine Medical Consultation)



2.2 Brainstorm & Idea Prioritization:-

Step-1: Team Gathering, Collaboration and Select the Problem Statement

1

Define your problem statement

PROBLEM

How might we make doctor appointment booking faster and more convenient for patients by providing real-time availability and an easy online scheduling system?

PROBLEM

How might we help doctors manage appointments efficiently through a centralized platform that organizes schedules and patient requests?

PROBLEM

How might we ensure a secure and trustworthy healthcare platform by implementing role-based access and doctor approval workflows?

PROBLEM

How might we allow patients to easily manage, reschedule, or cancel their appointments without needing manual assistance from hospital staff?

PROBLEM

How might we keep patients and doctors informed about appointment updates through a clear and reliable notification system?

Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

Narendra Mongam



3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

TIP
Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.

Patient Experience Ideas



Doctor Experience Ideas



Admin & Governance Ideas



Technical Feature Ideas



Step-3: Idea Prioritization

4

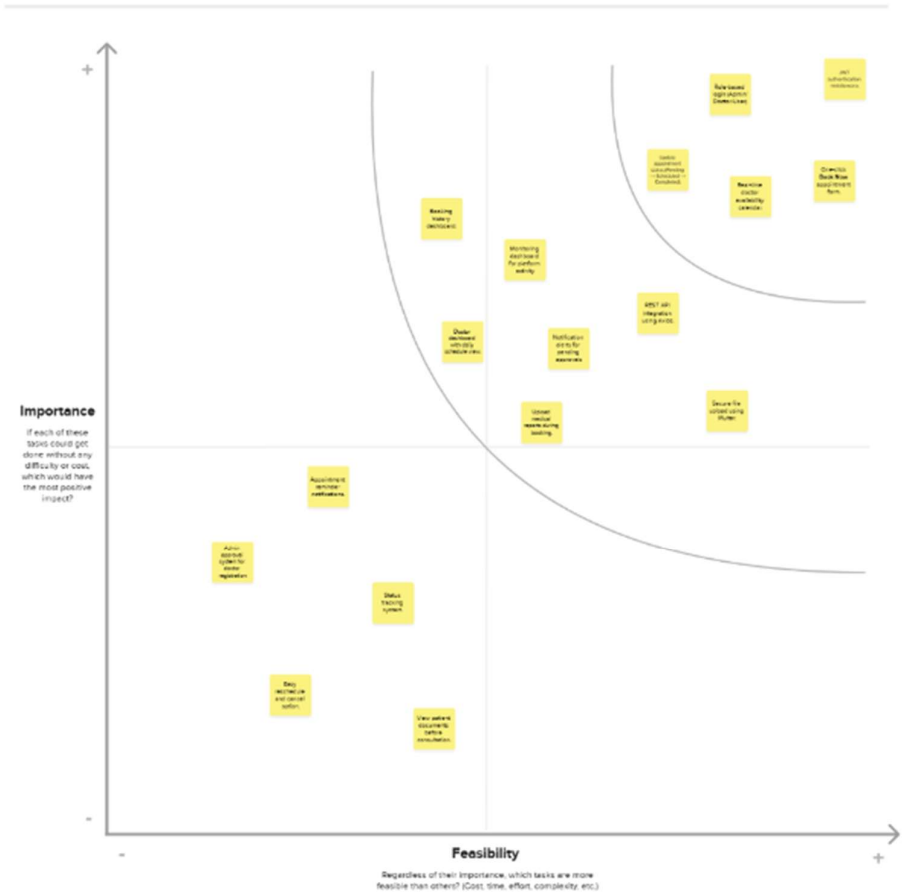
Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

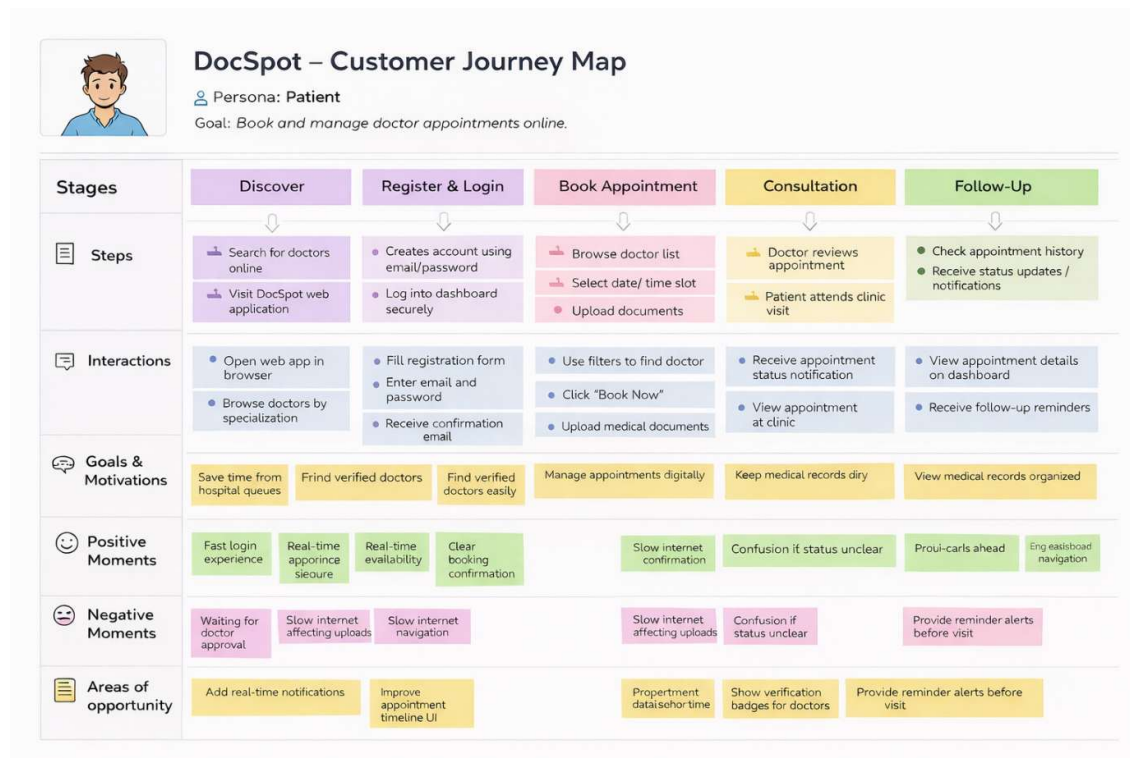
TIP

Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the laser pointer holding the H key on the keyboard.



REQUIREMENT ANALYSIS

3.1 Customer Journey map:



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Role selection (User / Doctor) Secure password creation
FR-2	User Authentication	Login using Email & Password JWT-based authentication
FR-3	Doctor Browsing & Search	View doctor list on dashboard Filter by specialization / availability
FR-4	Appointment Booking	Select date & time slot Upload medical documents Submit application request

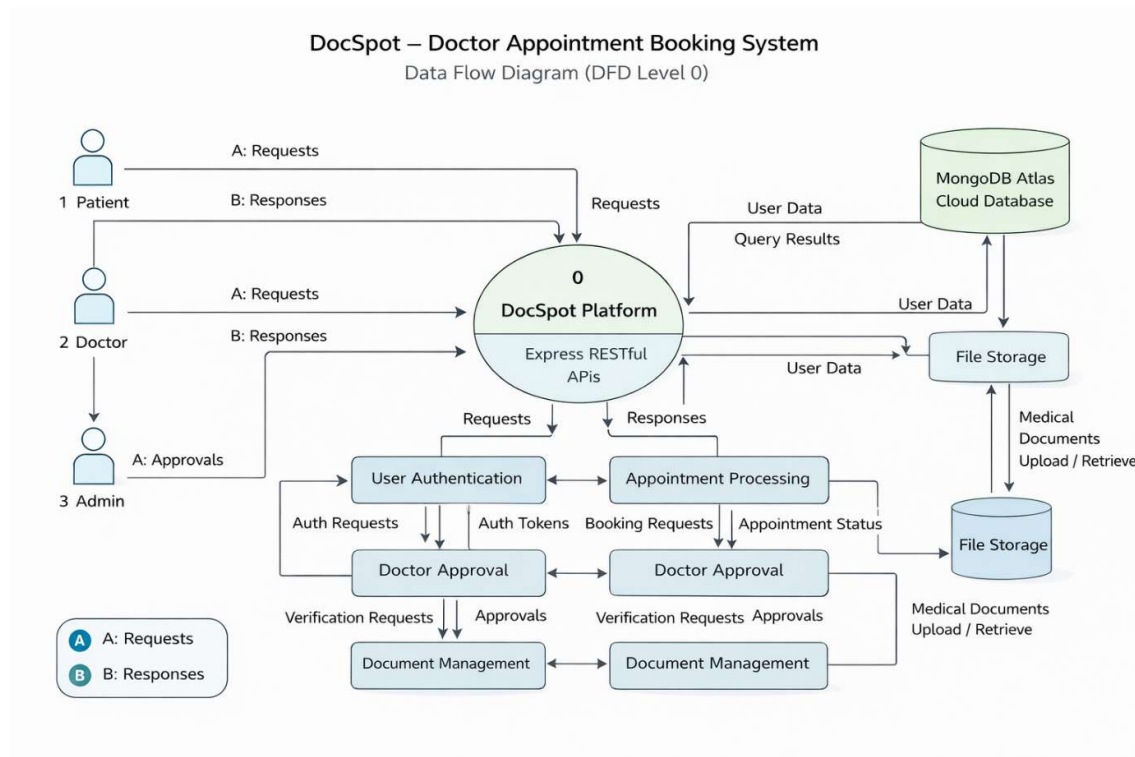
FR-5	Appointment Management	View booking history Cancel or reschedule appointment Track status (Pending / Scheduled / Completed)
FR-6	Doctor Dashboard	View incoming appointment requests Accept / Reject / Reschedule bookings
FR-7	Admin Governance	Approve doctor registrations Monitor platform activity
FR-8	Notification System	User receives confirmation updates Appointment status notifications

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple and user-friendly interface allowing patients to book appointments easily without technical knowledge.
NFR-2	Security	JWT authentication, encrypted passwords, and secure document upload to protect user data and privacy.
NFR-3	Reliability	System ensures accurate appointment scheduling and consistent status updates without data loss.
NFR-4	Performance	Fast loading of dashboards and real-time booking actions through efficient API communication.
NFR-5	Availability	Platform accessible online anytime for patients, doctors, and admins to manage appointments.
NFR-6	Scalability	MongoDB and REST architecture allow future expansion to more users, doctors, and healthcare services.

3.3 Data Flow Diagram: -



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web User)	Registration	USN-1	As a user, I can register using email and password	Account created & dashboard accessible	High	Sprint-1
Customer (Web User)	Authentication	USN-2	As a user, I can log in securely	Successful login redirects to dashboard	High	Sprint-1
Customer (Web User)	Doctor Listing	USN-3	As a user, I can view available doctors	Doctor list loads from database	High	Sprint-1
Customer (Web User)	Appointment Booking	USN-4	As a patient, I can book appointment with date & time	Appointment status shows "Pending"	High	Sprint-2
Customer (Web User)	Document Upload	USN-5	As a user, I can upload medical documents during booking	File uploads successfully	Medium	Sprint-2
Doctor	Appointment	USN-6	As a doctor, I can accept or reject appointment requests	Status updates to	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
	Management			Scheduled/Rejected		
Administrator	Doctor Approval	USN-7	As an admin, I can approve doctor registrations	Approved doctor visible in system	High	Sprint-3
Customer Care Executive	Notifications	USN-8	As a user, I receive appointment updates	Notification displayed in dashboard	Medium	Sprint-4
Customer (Web User)	Appointment History	USN-9	As a user, I can view booking history	Previous appointments visible	Medium	Sprint-5

3.4 Technology Stack: -

Technical Architecture: -

The DocSpot application follows a 3-Tier Client–Server Architecture:

- Presentation Layer (Frontend): React.js web interface where users, doctors, and admins interact with the system.
- Application Layer (Backend): Node.js and Express.js handle API requests, authentication, appointment processing, and role-based access control.
- Data Layer (Database): MongoDB stores user profiles, doctor details, appointment records, and notifications.

The frontend communicates with backend REST APIs using Axios, while JWT authentication secures protected routes. File uploads such as medical documents are handled through Multer middleware.

Table-1: Components & Technologies:

S. No	Component	Description	Technology
1.	User Interface	Web interface for patients, doctors, and admin dashboards	React.js, HTML, CSS, Bootstrap, Material UI
2.	Application Logic-1	Authentication & Role Management	Node.js, Express.js, JWT

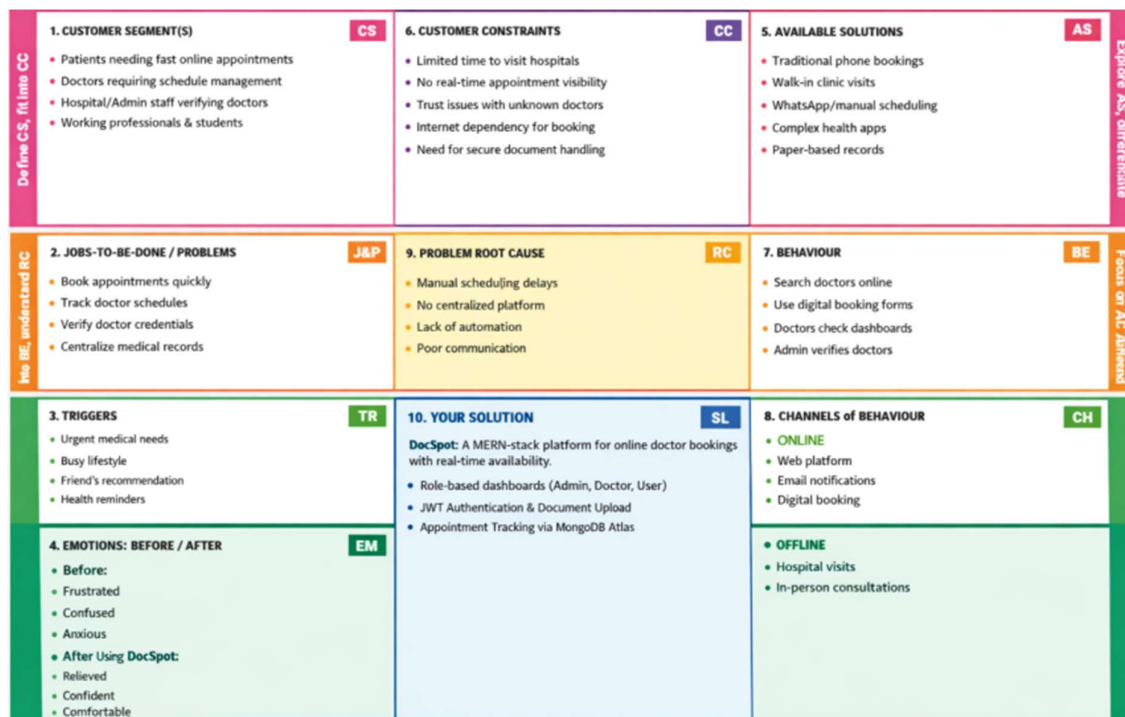
3.	Application Logic-2	Appointment Booking & Scheduling	Express.js REST APIs
4.	Application Logic-3	Notification & Status Management	Node.js Controllers
5.	Database	Stores users, doctors, appointments data	MongoDB, Mongoose
6.	Cloud Database	Cloud-hosted NoSQL database for storing users, doctors, appointments, and notifications with remote access and scalability	MongoDB Atlas
7.	File Storage	Medical document upload & storage	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	HTTP communication between frontend & backend	Axios
9.	External API-2	Not Applicable	—
10.	Machine Learning Model	Not used in this project	—
11.	Infrastructure (Server / Cloud)	Local development deployment	Node.js Local Server

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build UI & Backend APIs	React.js, Express.js, Node.js
2.	Security Implementations	Role-based authentication, encrypted passwords, protected routes	JWT, bcryptjs, Middleware
3.	Scalable Architecture	3-Tier architecture separating UI, backend logic, and database	REST Architecture, MongoDB
4.	Availability	Web application accessible anytime through browser	Node.js Server
5.	Performance	Fast API responses and asynchronous communication	Axios, Express.js

4. PROJECT DESIGN

4.1 Problem Solution Fit:



4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Patients face difficulty booking doctor appointments due to manual scheduling, lack of real-time availability, and long waiting times. Doctors need a centralized platform to manage appointments efficiently, while admins require a secure system to verify doctors and maintain platform trust.
2.	Idea / Solution description	DocSpot is a MERN-stack web application that enables patients to book appointments online through a centralized system. It provides role-based dashboards for Admin, Doctor, and User, real-time booking, secure JWT authentication, document upload, and appointment status tracking using REST APIs connected to MongoDB Atlas.
3.	Novelty / Uniqueness	Role-based healthcare workflow, admin-controlled doctor approval, real-time appointment tracking, secure medical

		document upload, and a simplified user-friendly booking interface.
4.	Social Impact / Customer Satisfaction	Reduces waiting time, improves healthcare accessibility, ensures verified doctor interactions, and enhances patient convenience through digital appointment management.
5.	Business Model (Revenue Model)	Commission-based booking model, subscription plans for doctors, premium doctor listing, and future telemedicine consultation services.
6.	Scalability of the Solution	Built using 3-tier architecture (React frontend, Express backend, MongoDB Atlas cloud database), enabling scalable performance and easy expansion to support more users and healthcare services.

4.3 Solution Architecture:

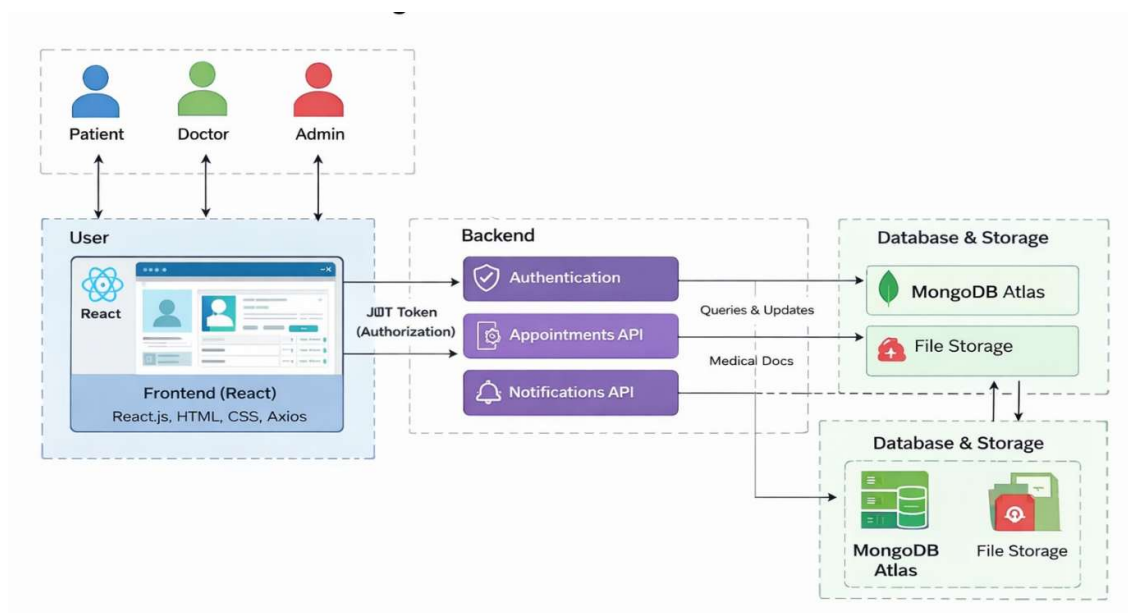


Figure-1: Architecture and data flow of the DocSpot doctor appointment booking platform

5. PROJECT PLANNING & SCHEDULING

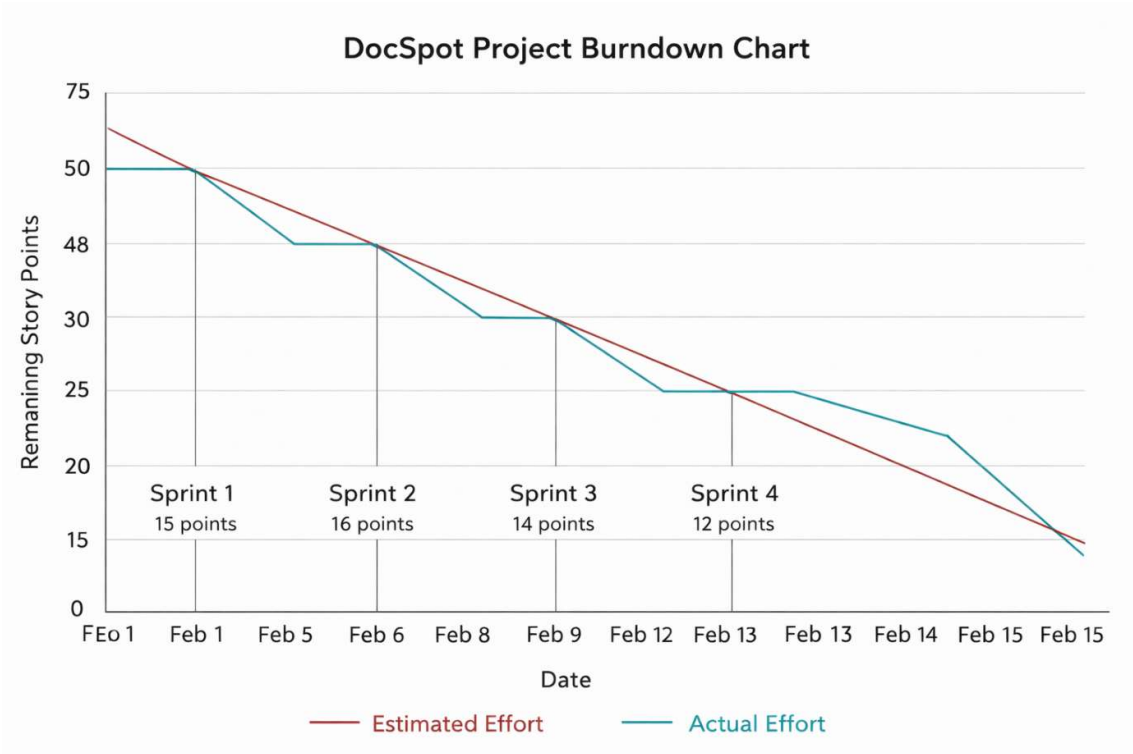
5.1 Project Planning:

Product Backlog, Sprint Schedule, and Estimation :

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register using email and password	2	High	Team
Sprint-1	Authentication	USN-2	As a user, I can log in securely using JWT authentication	2	High	Team
Sprint-1	Dashboard	USN-3	As a user, I can view doctor list on dashboard	3	High	Team
Sprint-2	Appointment Booking	USN-4	As a patient, I can book an appointment with date & time	5	High	Team
Sprint-2	Document Upload	USN-5	As a user, I can upload medical documents during booking	3	Medium	Team
Sprint-3	Doctor Panel	USN-6	As a doctor, I can accept or reject appointment requests	5	High	Team
Sprint-3	Admin Panel	USN-7	As an admin, I can approve doctor registrations	4	High	Team
Sprint-4	Notifications	USN-8	As a user, I receive appointment status updates	3	Medium	Team
Sprint-4	Appointment History	USN-9	As a user, I can view and manage booking history	3	Medium	Team

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	15	4 Days	01 Feb 2025	04 Feb 2025	15	04 Feb 2025
Sprint-2	16	4 Days	05 Feb 2025	08 Feb 2025	16	08 Feb 2025
Sprint-3	14	4 Days	09 Feb 2025	12 Feb 2025	14	12 Feb 2025
Sprint-4	12	3 Days	13 Feb 2025	15 Feb 2025	12	15 Feb 2025



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

Project Overview:

Project Name: DocSpot: Seamless Appointment Booking for Health

Project Description: DocSpot is a MERN-stack web application that enables patients to book doctor appointments online with real-time availability. The platform provides role-based dashboards for Admin, Doctor, and User, secure JWT authentication, document upload, and appointment status tracking using MongoDB Atlas.

Project Version: v1.0

Testing Period: 12 Feb 2025 to 15 Feb 2025

Testing Scope:

Features Tested:

- User Registration & Login
- Role-based Dashboard
- Appointment Booking
- Doctor Approval Workflow
- Document Upload
- Appointment Status Notifications

User Stories Tested:

- USN-1 Registration
- USN-2 Login
- USN-4 Appointment Booking
- USN-6 Doctor Approval
- USN-8 Notifications

Testing Environment:

URL/Location: http://localhost:5173

Credentials (if required): User → testuser@gmail.com / *****

Doctor → doctest@gmail.com / *****

Admin → admin@gmail.com / *****

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration	Step1: Open Register Page Step2: Enter details Step3: Submit	User account created successfully	Account created	Pass
TC-002	User Login	Step1: Enter email & password Step2: Click Login	Redirect to dashboard	Dashboard loaded	Pass
TC-003	View Doctors	Step1: Login Step2: Open Dashboard	Doctor list displayed	Doctors visible	Pass
TC-004	Book Appointment	Step1: Select doctor Step2: Choose date Step3: Submit	Appointment request created	Appointment pending	Pass
TC-005	Upload Document	Step1: Attach file during booking	File uploaded successfully	File stored	Pass
TC-006	Doctor Approval	Step1: Admin approves doctor	Doctor status updated	Approved successfully	Pass
TC-007	Appointment Status Update	Step1: Doctor accepts request	Status changes to Scheduled	Status updated	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Minor UI alignment issue in dashboard	Step-1: Login Step-2: Open Dashboard	Low	Closed	Fixed using CSS update
BG-002	File upload delay on slow internet	Upload large file	Medium	In Progress	Optimization suggested

Sign-off:

Tester Name: Mongam Narendra

Date: 15 February 2025

Signature: Mongam Narendra

7. RESULTS

7.1 Output Screenshots:

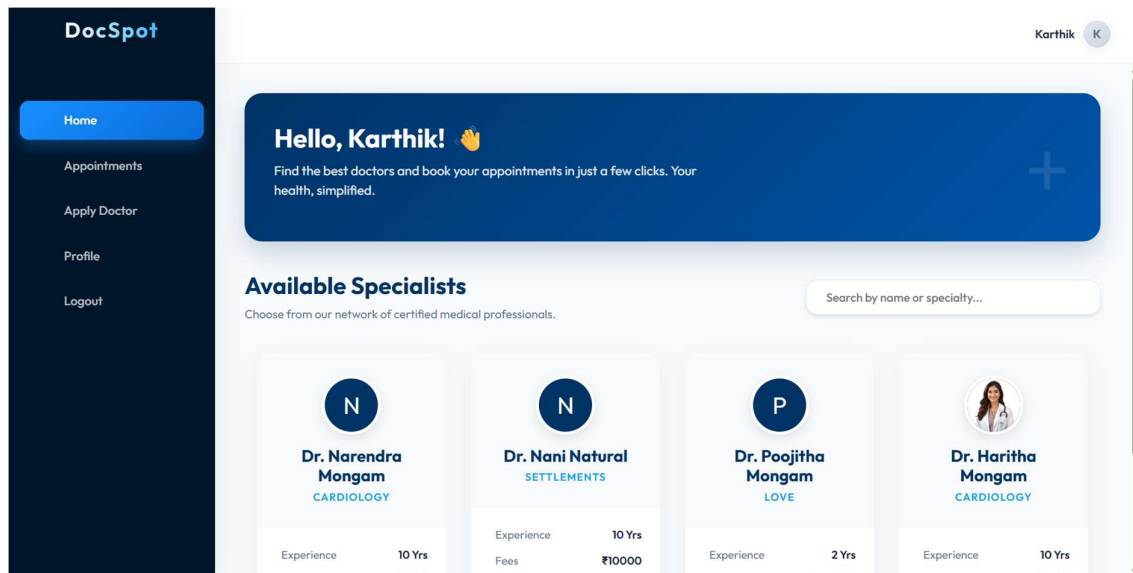


Figure-2: User Dashboard

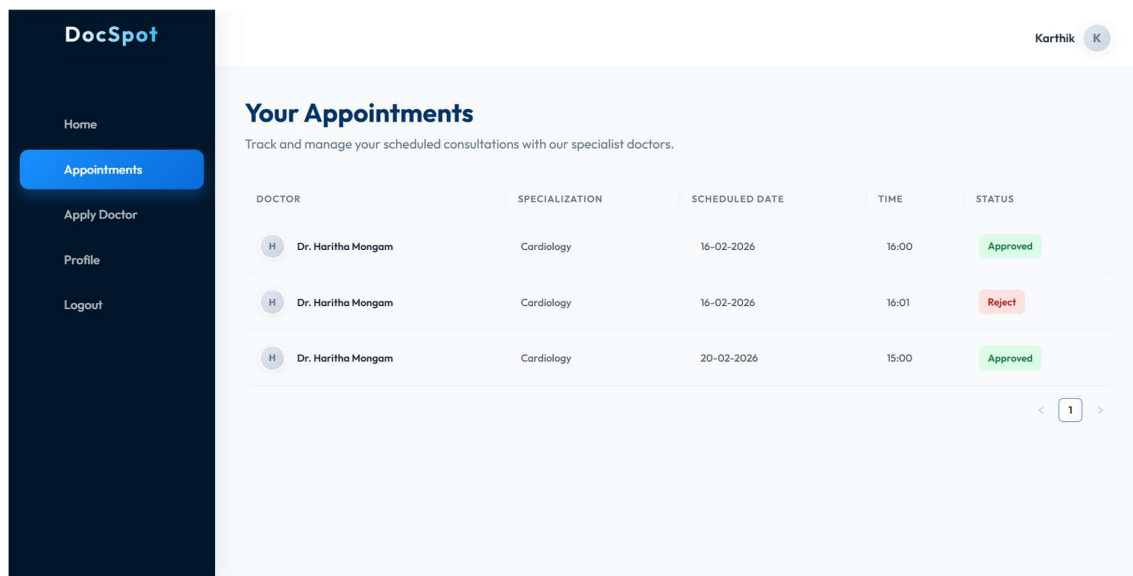


Figure-3: Appointment History

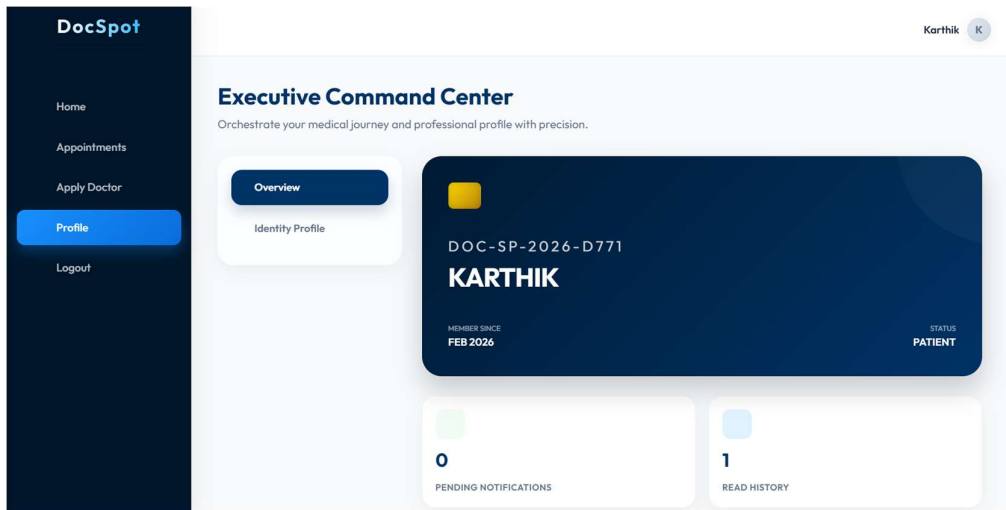


Figure-4: User Profile

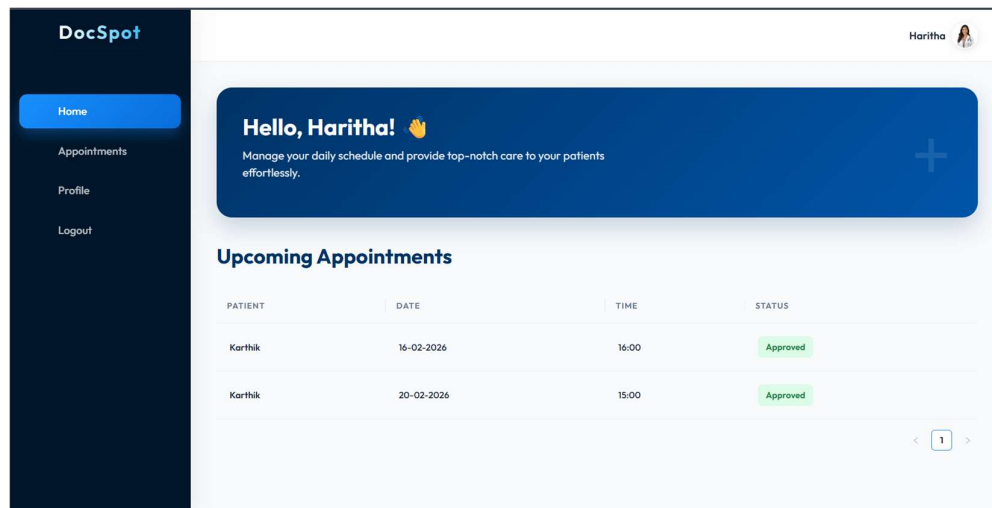


Figure-5: Doctor Dashboard

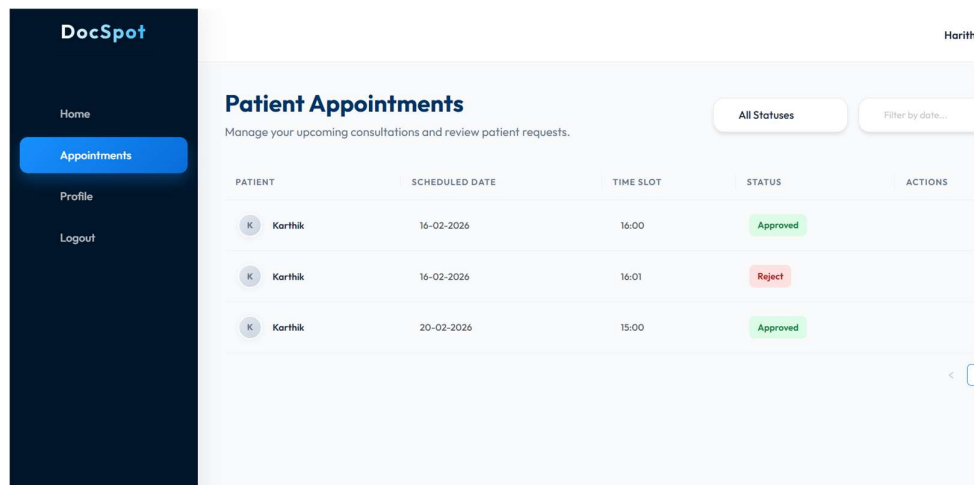


Figure-6: Doctor Dashboard (Appointment History)

DocSpot

Home

Appointments

Profile

Logout

Haritha

Specialist Profile

Manage your professional profile, availability, and consultation settings.

Professional Recognition

Professional Identity

First Name

Haritha

Last Name

Mangam

Contact Phone

9989375951

Professional Email

haritha@gmail.com

Medical Website

Clinic Address

123 Health St, Medical Center

Specialist Credentials

Specialization

Cardiology

Experience (Years)

10

Consultation Fee (₹)

500

Active Timings

12:00

18:00

Figure-7: Doctor Profile Page

DocSpot

Home

Doctors

Users

Profile

Logout

Admin User AU

All Doctors

Photo	Name	Status	phone	Actions
	Narendra Mangam	approved	9999999999	Authorized
	Nani Natural	approved	9999999999	Authorized
	Poojitha Mangam	approved	9999999999	Authorized
	Haritha Mangam	approved	9989375951	Authorized

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Figure-8: Admin Dashboard

DocSpot

Home

Doctors

Users

Profile

Logout

Admin User AU

Users List

Name	Email	Status	Doctor	Actions
narendra	narendra@gmail.com	active	Yes	Block
Test Doctor	testdoctor@example.com	active	No	Block
poojitha	poojitha@gmail.com	active	Yes	Block
Nani	nani@gmail.com	active	Yes	Block
Naren	mangam@gmail.com	active	No	Block
Haritha	haritha@gmail.com	active	Yes	Block
Karthik	karthik@gmail.com	active	No	Block

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Figure-8: Admin Can Block Users & Doctors

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8. ADVANTAGES & DISADVANTAGES

Advantages

- **Real-Time Appointment Booking:** Patients can view available doctors and book appointments instantly, reducing manual delays.
- **Role-Based Access Control:** Separate dashboards for Admin, Doctor, and User improve workflow management.
- **Cloud Database Integration:** Using MongoDB Atlas ensures scalability, data security, and remote access.
- **Secure Authentication:** JWT-based login protects user data and restricts unauthorized access.
- **Digital Document Management:** Patients can upload medical records during booking, improving consultation efficiency.
- **User-Friendly Interface:** Built with React and modern UI libraries for smooth interaction.

Disadvantages

- Requires stable internet connectivity to access the platform.
- File upload performance may depend on network speed.
- Currently supports only web-based access (no mobile app).
- Real-time notifications may require further optimization.

9. CONCLUSION

The DocSpot Doctor Appointment Booking Platform successfully addresses the challenges of traditional healthcare scheduling by providing a centralized digital solution. Through the use of modern web technologies such as React.js, Node.js, Express.js, and MongoDB Atlas, the system enables seamless interaction between patients, doctors, and administrators.

The platform improves healthcare accessibility by reducing waiting time, enabling efficient appointment management, and ensuring secure handling of medical data. The project demonstrates practical implementation of full-stack development concepts, RESTful APIs, authentication mechanisms, and cloud database integration.

10. FUTURE SCOPE

- Integration of **video consultation** for telemedicine services.
- Mobile application development using React Native.
- AI-based doctor recommendation system.
- Real-time push notifications using WebSockets or Firebase.
- Payment gateway integration for online consultation fees.
- Advanced analytics dashboard for doctors and administrators.
- Automated appointment reminders via SMS or Email.

12. APPENDIX

GitHub Repository:

<https://github.com/mongamnarendra/DocSpot-Seamless-Appointment-Booking-for-Health>

Live Demo / Project Video:

<https://drive.google.com/file/d/1pYC5oTLDDZh8cDte5w3fq4sR8zmdDD8h/view?usp=shareings>